Chapter 29

Measurement Uncertainty

The natural physical constants that define International System of Units (SI) base units of measurement have no associated measurement uncertainty (because they are, by definition, *natural physical constants*). Once humans become involved, these realized base units of measurement now have measurement uncertainty. The amount of measurement uncertainty can vary depending on the factors (components) discussed in this chapter. *Measurement uncertainty*, as defined in the *International Vocabulary of Metrology* (*VIM*) 2.26, is a "non-negative parameter characterizing the dispersion of the *quantity values* being attributed to a *measurand*, based on the information used."[1] In other words, the "true value" (i.e., "*quantity value* consistent with the definition of a *quantity*"[2]) is somewhere within its measurement uncertainty. We see this likelihood presented as measurement uncertainty $\pm x$ units at an approximately 95 % level of confidence where $k = 2$. In this chapter, we will be breaking down how this statement was formulated and what it means. We will begin by using definitions from the *VIM*.

It is important to begin by differentiating "measurement uncertainty" from "measurement error." They are not the same thing, nor are the terms interchangeable. Measurement error (error of measurement / error [*VIM* 2.16]) is defined as "*measured quantity value* minus a *reference quantity value*."[3]

- *Measured quantity value* / value of a measured quantity / measured value (*VIM* 2.10): **quantity value** representing a **measurement result**[4]
- *Reference quantity value* / *reference value* (*VIM* 5.18): **quantity value** used as a basis for comparison with values of **quantities** of the same **kind**[5]
- *Quantity value* / *value of a quantity* / *value* (*VIM* 1.19): number and reference together expressing magnitude of a **quantity**[6]
- *Measurement result* (*VIM* 2.9): set of **quantity values** being attributed to a **measurand** together with any other available relevant information[7]
- *Measurand* (*VIM* 2.3): **quantity** intended to be measured[8]
- *Quantity* (*VIM* 1.1): property of a phenomenon, body, or substance, where the property has a magnitude that can be expressed as a number and a reference[9]
- *Kind of quantity* / *kind* (*VIM* 1.2): aspect common to mutually comparable quantities[10]

- It estimates the dispersion of values associated with the measurement as a quantifiable numerical value.
- It provides a level of confidence in one's measurement.
- It is required to ensure metrological traceability.
- It must be accounted for in decision rules when making statements of conformity.
- It is a tool to help evaluate risk.
- It is used in validation and verification of methods.
- It is required by ISO/IEC 17025:2017:
  - Clause 6.4.5, "The equipment used for measurement shall be capable of achieving the measurement accuracy and/or measurement uncertainty required to provide a valid result."[21]
  - Clause 6.4.6, "Measuring equipment shall be calibrated when the measurement accuracy or measurement uncertainty affects the validity of the reported results."[22]
- It is required for laboratory accreditation by internationally recognized accreditation bodies.
- It is a good practice to identify and evaluate the potential impact of the many sources of error encountered when one performs a measurement.

MEASUREMENT UNCERTAINTY

How does measurement uncertainty happen? What are the contributing components? The "Notes" of the *VIM* definition of *measurement uncertainty* say: "Measurement uncertainty comprises, in general, many components. Some of these may be evaluated by **Type A evaluation of measurement uncertainty** from the statistical distribution of the quantity values from a series of **measurements** and can be characterized by standard deviations. The other components, which may be evaluated by **Type B evaluation of measurement uncertainty**, can also be characterized by standard deviations, evaluated from probability density functions based on experience or other information. Measurement uncertainty includes components arising from systematic effects, such as components associated with **corrections** and the assigned quantity values of **measurement standards**, as well as the **definitional uncertainty**. Sometimes estimated systematic effects are not corrected for but, instead, associated measurement uncertainty components are incorporated."[23]

Once measurement uncertainty is calculated, it must be evaluated. Measurement uncertainty is not static. For a given measurement or set of measurements, the measurement uncertainty is associated with a stated quantity value attributed to the measurand. When this value is changed, the associated measurement uncertainty also changes.[24]

EVALUATING MEASUREMENT UNCERTAINTY

When one makes any measurement, there is a measurement uncertainty associated with it. In an ideal world, the measurement that one makes would be absolute and have a true value associated with it; we would not have to worry about it because no measurement error would be present. This is described as the "one measurement bliss or one measurement quandary depending on whether one is an optimist or a pessimist."[25] Unfortunately, we do not live in an ideal world, and no measurement is perfect. These imperfections give rise to an error in the measurement result. An error has two components: random and systematic.[26] In this chapter we discuss what these components are and how to quantify them. Granted, there are situations in which one measurement is all that it is possible to make, even though multiple factors contribute to the uncertainty of measurement. That is why one should always consider estimating the uncertainty of measurement.

This section describes the process of estimating measurement uncertainty in a test and calibration environment. Practical examples using various parameters are used to illustrate the process of determining estimated measurement uncertainty.

MEASUREMENT UNCERTAINTY EVALUATION PROCESS

The process of evaluating measurement uncertainty can be done by following the *Guide to Uncertainty in Measurement* (*GUM*)[27] approach, as further clarified in NIST *SOP 29* (*SOP for the Assignment of Uncertainty*)[28]:

1. Specify the measurand by specifying the measurement process.
2. Identify all possible sources of measurement uncertainty and characterize the measurement uncertainty components.
3. Quantify each measurement uncertainty component/contributor in applicable measurement units.
4. Convert measurement uncertainty components to standard uncertainties in the units of the measurement result.
5. Calculate the combined uncertainty by combining the contributors (often by the root sum square [RSS] method).
6. Expand the combined uncertainty using an appropriate coverage factor (k factor).
    - Note: There is a common misconception that the coverage factor is $k = 2$. This is only true if the overall effective degrees of freedom is infinite at approximately 95 % confidence interval. At other effective degrees of freedom, the k coverage factor may be different.
7. Review the entries and evaluate the expanded measurement uncertainty against appropriate tolerances, user requirements, and laboratory capabilities.
8. Report correctly rounded (two significant digits) measurement uncertainties with associated measurement results in a measurement

OTHER MEASUREMENT UNCERTAINTY CONSIDERATIONS

The Effect of Resolution (Readability, Sensitivity) on Measurement Uncertainty

ILAC P14:09/2020 addresses when the resolution of the inspection, measurement, and test equipment (IM&TE) being calibrated (UUT) should be included by stating, "When it is possible that the best existing device can have a contribution to [measurement] uncertainty from repeatability equal to zero, this value may be used in the Evaluation of the CMC. However, other fixed uncertainties associated with the best existing device shall be included."[43]

One cannot report the calibration measurement uncertainty for any UUT at the same value as the Calibration and Measurement Capability (CMC) uncertainty component, unless the UUT has the same performance and specifications as the "best existing device." "Contributions to the [measurement] uncertainty stated [on the calibration certificate] shall include relevant short-term contributions during calibration and contributions [such as UUT resolution] that can reasonably be attributed to the customer's device. Where applicable the [measurement] uncertainty shall cover the same contributions to uncertainty that were included in evaluation of the CMC uncertainty component, except that [measurement] uncertainty components evaluated for the best existing device shall be replaced with those of the customer's device. Therefore, reported [measurement] uncertainties tend to be larger than the [measurement] uncertainty covered by the CMC."[44]

At a minimum, the expanded measurement uncertainty should include the uncertainty of the measurement process (labeled as CMC, though it is the CMC uncertainty component), as well as the UUT's resolution. There are some instances in which the UUT's repeatability is substituted with that of the best existing device used for calibration, as referenced in *ILAC P14:09/2020*. As the resolution of the device increases, so does the overall measurement uncertainty. Not accounting for the UUT's resolution can result in an increased risk to the end user unless the same resolution was accounted for in the CMC uncertainty component.[45]

Sensitivity Coefficients[46]

Sensitivity coefficients are defined as "the differential change in the output estimate generated by a differential change in the input estimate divided by the change in that input estimate." These derivatives describe how the output estimate (y) varies with changes in the values of the input estimates ($x_1, x_2, \ldots, x_n$).

Sensitivity coefficients show how components are related to results. The sensitivity coefficient shows the relationship of the individual (measurement) uncertainty component to the standard deviation of the reported value for a test item. Most sensitivity coefficient values for Type B evaluations will be "1," with a few exceptions.

Effective Degrees of Freedom[47]

According to K. A. Brownlee, "Degrees of freedom for the standard uncertainty, u, which may be a combination of many standard deviations, is not generally known.

$$v_{\text{eff}} = \frac{u_{\text{c}}^4(y)}{\sum_{i=1}^{N} \frac{u_i^4(y)}{v_i}}$$

Figure 29.12 Welch-Satterthwaite formula.

This is particularly troublesome if there are large components of uncertainty with small degrees of freedom. In this case, the degrees of freedom are approximated by the Welch-Satterthwaite formula,"[48] as seen in Figure 29.12.

where

v_i = Degrees of freedom
u_c = Combined [measurement] uncertainty
u_i = Individual uncertainties
V_{eff} = Effective degrees of freedom

For further information, refer to *JCGM 100,* Annex G, Section 4.

The *t*-Distribution and Degrees of Freedom

For instances where the effective degrees of freedom are small, one needs to use the *t*-distribution (Student's distribution) to determine the *k* value for the coverage factor in the expanded combined (measurement) uncertainty. Both the *GUM* (*JCGM 100:2008*)[49] and the NIST/SEMATECH *e-Handbook of Statistical Methods*, 1.3.6.7.2. Critical Values of the Student's *t*-distribution,[50] further explain how and when to use Student's *t*-distributions. See Figure 29.13 for the Student's *t*-distribution table.

MANAGING MEASUREMENT UNCERTAINTY

It is important to ensure that once the measurement uncertainty for a process (calibration or test) has been determined, the data are not simply filed away and forgotten. The work is not finished. Managing measurement uncertainty is a continuous process, not a one-time exercise. When a manufacturer provides measurement uncertainty data for its product, they are specified under stated operating conditions. The manufacturer may specify 30-day, 90-day, one-year, or other periods for use under defined humidity and temperature conditions.

It is important to estimate measurement uncertainty under the laboratory's operating environment and not the manufacturer's stated conditions. From time to time, the measurement uncertainty estimates should be reevaluated to ensure that estimates have not significantly changed. At a minimum, whenever equipment is recalibrated or readjusted, an evaluation is necessary to ensure that its measurement uncertainty has not changed.

Various tools exist for managing measurement uncertainty data. Use of software minimizes calculation errors and helps in managing data from a computer workstation. Computer spreadsheet packages also help in easing data calculations.